FIGURE 2.1 Simulated random walk.
Source: Authors.

FIGURE 2.2 Simulated mean-reverting process.
Source: Authors.

in this example. In fact, the two series were constructed with identical normal random variates. In the case of the random walk, the mean of each observation was the value of the previous observation, so that the process was a martingale. In the case of the mean-reverting process, the mean of each observation was set to reflect the tendency for the process to return to the mean. At this point, we'd hope most readers would identify Figure 2.2 as the one with the mean-reverting variable. If we observe both figures closely, we can see that the mean-reverting process is in some sense a transformation of the random walk in Figure 2.1.

The speed with which a variable tends to revert toward its mean can vary. For example, Figure 2.3 and Figure 2.4 show time series that were simulated using the same random normal variates that generated the mean-reverting variable in Figure 2.2 but with an important difference. The variable in

FIGURE 2.3 Simulated mean-reverting process: faster mean reversion.
Source: Authors.

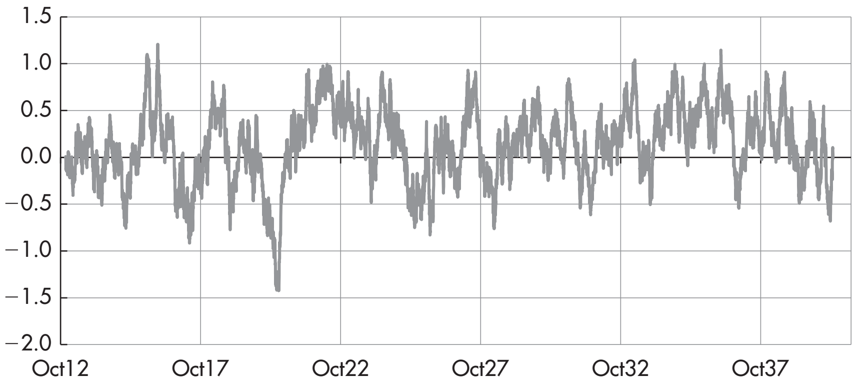

FIGURE 2.4 Simulated mean-reverting process: even faster mean reversion.
Source: Authors.

Figure 2.3 was constructed to have a faster speed of mean reversion than the variable in Figure 2.2, while the variable in Figure 2.4 was constructed to have a still faster speed of mean reversion.

While it's well and good to consider variables simulated via known equations by a computer, traders and analysts have to make judgments about real-world data, which are almost always messier in some respects than simulated data. So it's also useful to consider a few real-world examples.

Figure 2.5 shows the spot price of gold in US dollars since January 1975. In our view, the strong upward drift exhibited in this series makes it a poor candidate to be modeled by a mean-reverting process.

Figure 2.6 shows the realized volatility of the ten-year (10Y) US Treasury yield since January 1962. Given that this series has repeatedly returned to a

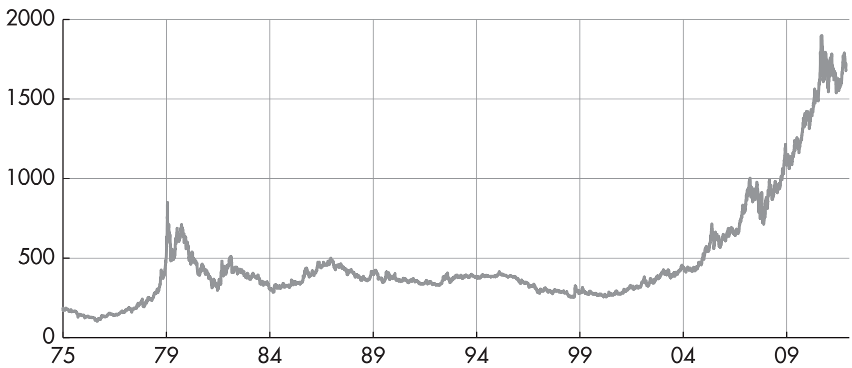

FIGURE 2.5 Spot price of gold in US dollars since January 1975.
Source: data – Bloomberg, chart – Authors.

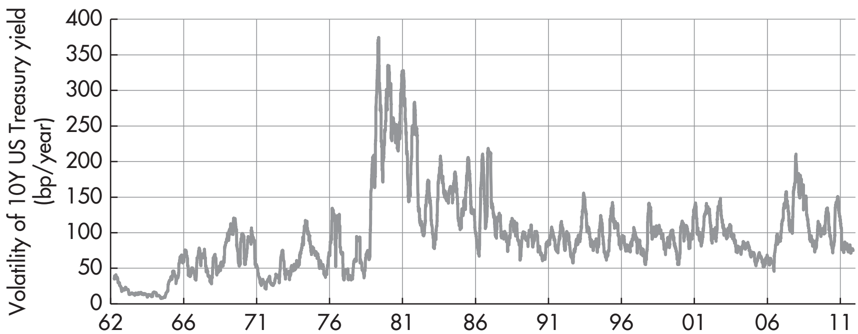

FIGURE 2.6 Realized volatility of 10Y US Treasury bond yield (bp/year).
Source: data – Bloomberg, chart – Authors.

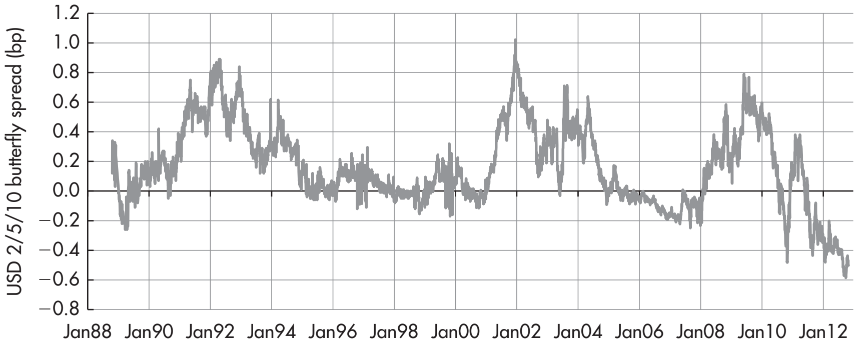

FIGURE 2.7 2/5/10 butterfly spread along USD swap curve since 1988.
Source: data – Bloomberg, chart – Authors.

long-run mean in the past, it appears to be a relatively good candidate for modeling with a mean-reverting process.

As another example, Figure 2.7 shows the 2Y/5Y/10Y butterfly spread along the USD swap curve since 1988. Given the number of times during the sample that this spread returns to its long-run mean, we consider it another good candidate for modeling with a mean-reverting process.

Mathematical Definitions

Having provided verbal and graphical intuition regarding mean reversion, it's time to attempt to provide a few useful mathematical definitions.

Stochastic Differential Equation

First, we'll provide a brief definition of a *stochastic differential equation* (SDE). In practice, this term is fairly simple to define, as most of the definition is contained within the name. In other words, it's an equation that characterizes the random behavior of a variable over an infinitesimal period of time. As a result, it gives us the data-generating mechanism for the variable. For example, the equation would allow us to simulate the variable over time using a computer.

For example, $dx_t = k(\mu - x_t)dt + \sigma dW_t$ is the SDE for an Ornstein–Uhlenbeck (OU) process, the continuous-time limit of a first-order autoregressive process. The OU process is a popular SDE for modeling mean-reverting variables, as it has moments and densities that can be expressed analytically. In this equation, dx_t is the change in the value of the random variable x at time t, over the infinitesimal interval dt. The speed of mean reversion is given by the parameter k and the long-run mean of the variable is given by μ.

The instantaneous volatility of the variable is given by σ, and the term dW_t is the change in the value of W_t over the instantaneous time interval dt. In fact, W_t is ultimately the source of randomness that drives the process in this equation. In particular, W_t is a pure random walk, often referred to as Gaussian white noise. W_t is also referred to as a Wiener process, after the American mathematician Norbert Wiener.

In general, SDEs take the form

$$dx_t = f(x_t)dt + g(x_t)dW_t$$

The term $f(x_t)$ is the drift coefficient of the equation, and it defines the mean of the process. The term $g(x_t)$ is the diffusion coefficient of the equation, and it defines the volatility of the process.

Linking mathematics with intuition, one can think of this SDE as separating the driving forces of a trade into a non-stochastic part (there is no randomness in the drift term $f(x_t)dt$) and into a stochastic part (the diffusion coefficient $g(x_t)dW_t$). Depending on the choice of the functional form for the drift coefficient, it may contain other non-stochastic elements besides the mean reversion as well. Hence, the drift coefficient can be interpreted as containing the non-stochastic, therefore predictable, driving forces of a trade, i.e. as a quantification of its return predictability. In contrast, the diffusion coefficient can be associated with the stochastic, unpredictable risk in a trade. Together, the SDE can be thought of as separating and connecting return and risk. As a consequence, expressing a trade via such an SDE is a suitable way to calculate its relative performance and risk characteristics, such as a Sharpe ratio. We will illustrate this calculation with an example below.

Analysts familiar with regressions may find the perspective on mean reversion depicted in Figure 2.8 intuitive. Using the data of the example shown in Figure 2.18 and their discrete steps, in this case, trading days, each point represents the move from a given day to the next, i.e. $x_{t+1} - x_t$, as a function of the (negative) distance from the mean on that given day, i.e. $\mu - x_t$. The resulting cloud of points shows the daily moves on the y-axis as depending on the (negative) distance from the mean on the x-axis. One can now fit a linear regression line through these observations. The slope of this linear regression line is linked to the speed of linear mean reversion, i.e. the parameter k of an OU process. In fact, the slope of the regression line (0.055) is very close to the speed of mean reversion (0.056) of an OU process fitted to the same data. And adding other non-stochastic elements besides the mean reversion to the drift term would correspond to moving to a multiple regression.

Conditional Density

Next, we'll define the *conditional density* of a process, also referred to as a *transition density*. In particular, the conditional density gives us the probability

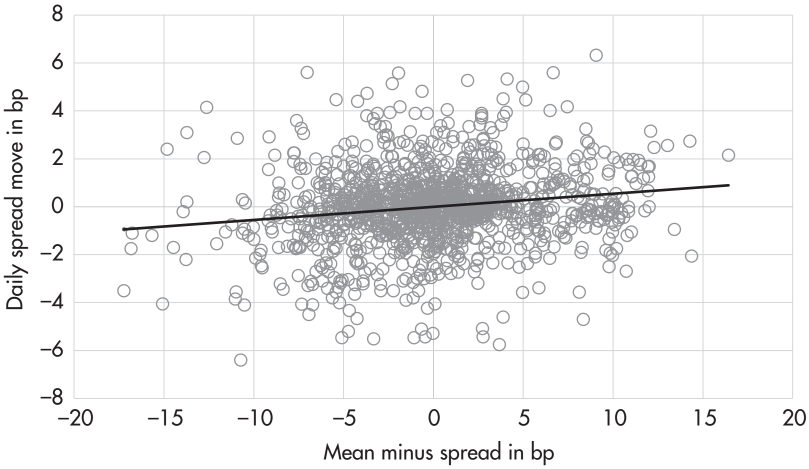

FIGURE 2.8 Daily moves of the swaption volatility difference EUR 5Y5Y – GBP 5Y5Y as a function of the (negative) distance to the mean.
Source: Authors.

density for the future value of a random variable conditional on knowing some other information about the variable. In the case of a time series process, the conditioning information is usually some earlier value of the variable. For example, in the case of the OU process, the transition density of $x_{t+\tau}$ for $\tau > 0$ is a normal density with mean given by $\mu + (x_t - \mu)e^{-k\tau}$ and with a variance given by $\frac{\sigma^2[1-e^{-2k\tau}]}{2k}$.

Unconditional Density

The unconditional density of a process is the probability density for the future value of a random variable without being able to condition the density on any additional information. You could think of the unconditional density as the histogram that would result from simulating the process over an infinitely long period. More precisely, it's the limit of the conditional density $p(x_{t+\tau})$ as τ goes to infinity. So in the case of the OU process, the unconditional density is a normal density with mean given by μ and with variance given by $\frac{\sigma^2}{2k}$.

Stationary Densities and Mean-Reverting Processes

In some cases, a variable will have a conditional density, but it won't have an unconditional density. In other words, the limit of the conditional density $p(x_t)$ won't converge to a limiting density.

A simple example of this would be a random walk with drift, given by the SDE $dx_t = \rho dt + \sigma dW_t$. The transition density or unconditional density for this process is normal with mean $x_t + \rho\tau$ and variance given by $\sigma^2\tau$. In this case, neither the mean nor the variance has a limit as $\tau \to \infty$, and the limit of the conditional density doesn't exist.

Even in the case of a random walk with no drift, given by the SDE $dx_t = \sigma dW_t$, there is no unconditional density. In this case, the mean for all future transition densities is simply the current value of the variable x_t, but since there is no limit to the variance of the process, there is no limit to the conditional density, and the unconditional density doesn't exist.

However, in many cases, the limit of the conditional density will exist, and the random variable is said to be mean reverting. We also say that stationary density exists and that the process is stationary.

Return Predictability and Alpha

Having provided some intuition and some mathematical definitions of mean-reverting processes, it's helpful to take a step back and consider the usefulness of mean-reverting models for investors and traders.

Return predictability is a necessary, though not sufficient, condition for generating alpha, defined here as an atypically high, risk-adjusted return. If we identify a financial variable that exhibits return predictability, then either the risks of that variable are predictable or the risk-adjusted returns are predictable.

Mean reversion is a form of return predictability. If a financial variable exhibits mean reversion, then we can use that information to improve our predictions for the future value of the variable. In our view, more often than not, mean reversion in a financial variable is an indication that risk-adjusted returns of the variable are predictable. Of course, in some cases, some or all of the mean reversion in a variable will be the result of risks that exhibit mean reversion rather than the result of mean reversion in the risk-adjusted returns. But, in our view, the more typical result is that the risk-adjusted returns are predictable. In this case, mean reversion can be used to generate alpha for traders and investors.

DIAGNOSTICS FOR MODEL SELECTION

The key for modeling any mean reversion in the variable x is to select a functional form for the drift coefficient, $f(x)$, that is useful in depicting the tendency of x to decline toward its long-run mean when it's above the mean and to increase toward its long-run mean when it's below the mean. So at a minimum, we need a function $f(x)$ that satisfies three conditions:

- The value of $f(x)$ is negative when x is above the long-run mean.
- The value of $f(x)$ is positive when x is below the long-run mean.
- The value of $f(x)$ is zero when x is equal to the long-run mean.

Of course, one simple function that satisfies these properties is a line, in which $f(x)$ could be parameterized as $f(x) = k(\mu - x)$. In this case, μ is the long-run mean of the process and k is the strength with which the variable x is "pulled" toward the long-run mean. An equivalent parameterization of $f(x)$ is $f(x) = a + bx$, in which case, $a = k\mu$, and $b = k$.

As it happens, the simplicity of this linear parameterization simplifies the estimation of parameters from historical data, as the likelihood function will often have a closed-form representation in this case, depending on the specification of the diffusion coefficient, $g(x)$. The linear specification for $f(x)$ also simplifies the calculation of transition densities and first passage time densities for x.

But of course a line is not the only function that could be used to represent the mean-reverting tendencies of x, and we may be willing to sacrifice some simplicity in exchange for a functional form for the drift coefficient that is more useful in capturing the actual mean-reverting tendencies exhibited in the data.

For example, a more flexible functional form that has been used in a variety of applications is $f(x) = a + bx + cx^2 + dx^3$, a third-order polynomial in x. In particular, this nonlinear specification allows for the variable x to exhibit increasingly strong mean reversion as it moves further away from the long-run mean. A similar function form that has been used successfully in a variety of applications is $f(x) = \frac{a}{x} + bx + cx^2 + dx^3$, as the $\frac{1}{x}$ term allows for the drift coefficient to become increasingly strong as x approaches zero, allowing zero to act as a reflecting barrier for the process. Examples of these three functional forms appear in Figure 2.9.

In this case, we want to restrict the value of d to be non-positive, to avoid the drift coefficient going to infinity as x increases, in which case x would be an explosive process rather than a stationary, mean-reverting process. For a similar reason, we'd like to restrict the value of a to be non-negative.

Strictly speaking, the drift coefficient, $f(x)$ also needs to satisfy other, rather technical, mathematical conditions in order to ensure that this SDE has a solution. In practice, we tend to assume that these conditions are satisfied (perhaps more often than we should), as the processes typically studied in financial applications tend to be well behaved.

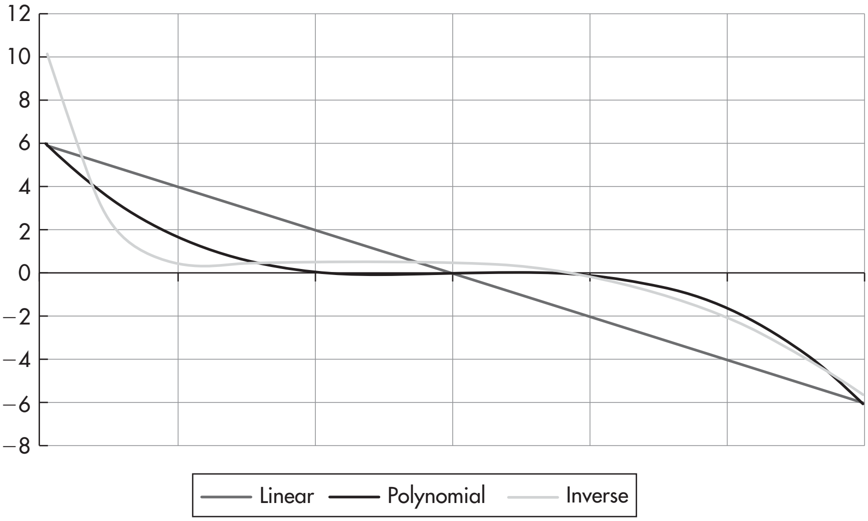

FIGURE 2.9 Examples of drift coefficients.
Source: Authors.

At this point, we should stress that the drift coefficient, $f(x)$, can assume an unlimited variety of functional forms, including nonparametric forms, and the particular form used in practical applications must be specified by the analyst.

If the analyst is open to sacrificing some analytical tractability in an attempt to more usefully model certain aspects of the process, he would benefit from some diagnostic guidance as to the functional forms for $f(x)$ that are likely to be most useful.

To our knowledge, the most useful diagnostic is one suggested by Richard Stanton in a Stanford research paper in the mid-1990s. The basic idea behind this diagnostic tool is to create a nonparametric, empirical approximation of the drift coefficient using historical data. An illustration of this is provided in Figure 2.10.

Each point in the graph represents the estimated strength of mean reversion when the variable assumes values in the neighborhood of that point. Perhaps the most expedient way to explain this concept is to start by listing the steps involved in creating the graph.

1. Group the observations into “buckets,” with the number of buckets determined by the analyst.
2. For each bucket, calculate the average subsequent change of each observation in the bucket.

3. For each bucket, plot a point with a horizontal coordinate equal to the midpoint of the observations in the bucket and with a vertical coordinate equal to the average subsequent change of the observations in the bucket.

For example, let's imagine that we have 1,000 observations in our data series and that we want to group these into 20 buckets of equal width, with equally spaced midpoints. We can separate the range (high–low) into 20 segments of equal length and then place each of our 1,000 observations into one of these 20 buckets. Some buckets may have many observations, while some buckets may have relatively few observations. As a general rule, if a bucket has very few observations, it would be wise to decrease the number of buckets, which should increase the number of observations in most buckets, in an attempt to reduce the estimation error of the average change within that bucket.

Once each observation has been placed into a bucket, calculate the subsequent change of each observation in the bucket. For example, if the 400th data point in the series has been placed into the sixth bucket, the subsequent change associated with the 400th observation would be the 401st observation less the 400th observation, even if the 401st observation is itself within a different bucket.

In other words, we're calculating the average change of observations whose starting values are within the same bucket, and then we repeat the calculation for each of the remaining buckets.

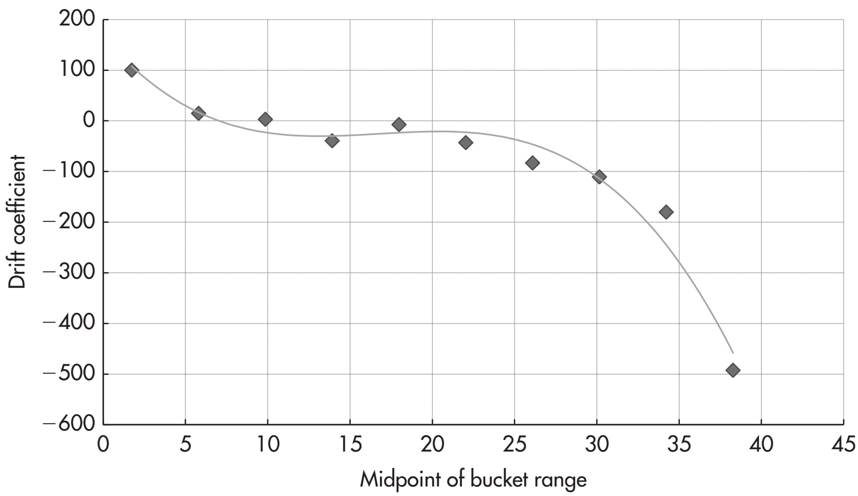

FIGURE 2.10 Diagnostic tool for drift coefficient.
Source: Authors.